

PulseGuard — a multi-event intraoperative early-warning system, and a study of when it can be trusted

VitalDB (5,151 surgeries) · zero-shot external validation on MOVER / UC-Irvine · demo: aastikrajan.github.io/pulseguard · code + benchmark: github.com/AastikRajan/pulseguard

Research prototype, not clinical software. All metrics are out-of-sample on a single locked by-case split, or on a second hospital with no retraining.

A streaming model scores, second by second during surgery, the probability that a patient will deteriorate within the next five minutes — across nine event types — together with a body of work establishing **where that signal is real, where it generalizes to another hospital, and where it is causal rather than confounded**. It was built solo on public data. The model is the small part: most of the effort went into a few dozen controlled experiments on trust — calibration, distribution shift, leakage, and causal identification.

Setup and evaluation

- **Data:** VitalDB (Seoul National University Hospital) — 5,151 surgeries, 35 signals at 1 Hz (plus 500 Hz waveforms), 1.57M windows, ~105k labeled events.
- **Split:** 20% of cases held out by patient (fixed seed) = **1,030 cases / 303,909 windows**; no case in both train and test; nothing tuned on the test set.
- **External:** zero-shot validation on MOVER (UC-Irvine), 19,019 surgeries — the VitalDB model applied with no fine-tuning.

What it does (held-out, and external where noted)

Capability	Result
Unified early-warning — any deterioration, 5-min horizon	AUPRC 0.454 · AUROC 0.849 · ~4.3-min median lead (base rate 12%)
Alarm precision after conformal calibration	0.62–0.71 vs ~0.12 for deployed monitors; ~300× fewer alarms
Multi-event detection — 9 events, independent detectors	per-event AUROC 0.74–0.96
Zero-shot cross-hospital (Seoul → UC-Irvine, no retraining)	AUROC 0.78–0.80
Causal: does hypotension cause post-op AKI?	RR 1.77 → 1.07 (g-formula; ~90% confounding)

What the experiments established

- **A transportability principle.** Only universally-dense channels (heart rate, SpO₂) transfer across hospitals; adding rich source-specific channels (invasive pressure, the ETCO₂/tidal-volume cascade) improves in-hospital AUPRC but consistently *degrades* zero-shot transfer — shown three independent ways. (The literature calls cross-continental transfer of these models “virtually untested.”)
- **Leak-free event-history features** (recurrence, cascade, time-since-induction) add **+0.011 AUPRC at zero inference cost**, derived from the event log the model already keeps.
- **Architecture:** nine events are best handled by independent per-event detectors — a single 9-way differential degrades the original six because events co-occur. Light anesthesia / awareness is detectable (AUROC 0.93).
- **Causal arm:** the hypotension→AKI association is ~90% confounding by indication (deconfounded with a g-formula, E-value and negative-control bounded), **replicated on the second hospital**, and consistent with the 2025 null prevention trials.
- Two reproducible hypoxemia phenotypes (disconnect/apnea vs atelectasis); a validated **3-minute-early circuit-integrity alert** at <1% false-alarm rate.

How it works

- 30-second windows → features (levels, short-term trends, variability, cross-signal coupling, event-history) → a gradient-boosted ensemble scoring 5-minute deterioration risk.
- Scores become trustworthy alarms via **conformal calibration** (a precision target), then **episode-grouping** collapses bursts into single alerts — the $\sim 300\times$ reduction.

Where it can't be trusted — the boundary

- **Intraoperative hypotension is close to unpredictable** from the monitor (recall ~ 0.12). I established this **seven independent ways** (operating-point, lead-signal tests, critical-slowness, a dynamical-network biomarker, short-horizon reachability, phenotype homogeneity, and 500 Hz heart-rate/blood-pressure variability) and report it as a hard limit rather than tuning around it.
- Four channels carry nearly all the signal (heart rate, SpO₂, the pressure line, the ventilator); heart rate and SpO₂ have no proxy — a low-cost ECG + pulse-oximetry version is plausible where arterial lines aren't available.

On rigor, and what I'm looking for

One locked by-case split, chosen once. A data leak in my own pipeline that had inflated the headline ($0.48 \rightarrow 0.454$) was caught and quarantined; controlled re-tests overturned several of my own earlier claims; every null is reported. I haven't found a published system that combines multi-event early-warning, zero-shot cross-hospital transfer, a causal analysis, and calibrated alarms alongside an explicit map of what it cannot predict. **I'm looking for feedback, a research role, or a collaborator** to take the strongest pieces — the cross-hospital transfer and the honest benchmark — toward a paper and prospective external validation.